DEPI · PROFESSIONAL DATA ANALYSIS

DateSight

Requirements Gathering — Stakeholders · User Stories & Use Cases · Functional & Non-functional Requirements

Item	Detail
Project	DateSight – Egyptian Dates Export Analytics & Dashboards
Programme / Track	DEPI – Digital Egypt Pioneers Initiative · Professional Data Analysis
Submission item	3. Requirements Gathering
Team	5 members: 1 BIS · 2 GIS · 1 Agriculture · 1 MIS
Version / Date	v1.0 · October 2026
Repository	GitHub (link to be added by the project lead)

1. Introduction

Purpose. This document captures what DateSight must do and how well, from the point of view of the people who will use or depend on it. It feeds the System Analysis & Design milestone (6 Nov 2026).

Product. Interactive Tableau dashboards on Egypt's dates exports (HS 080410), 2016–2025, supported by a documented, validated data model.

How requirements were gathered. Analysis of the provided export dataset and of public trade reports and platforms; review of similar export dashboards; team workshops across the five backgrounds (BIS, GIS, Agriculture, MIS); DEPI project guidelines. Needs of external stakeholders are *derived from desk research* and will be validated with real users during user acceptance testing.

> **Priority scale (MoSCoW):** Must = required for acceptance · Should = important, included if time allows · Could = nice to have. **Conditional** = depends on obtaining the necessary data (country-level exports, quantities, varieties).

2. Stakeholder Analysis

Stakeholder	Type	Needs and expectations	Influence	Interest	How we engage
Egyptian dates exporters	Primary user	Trend and growth of exports, best markets, competitors, price benchmarks, opportunities	Medium	High	Design the dashboard around their questions; test with exporters or export-minded users
Ministry of Agriculture and Land Reclamation	Primary user / policy	National export performance, market diversification, gap between production and exports	High	High	Executive overview and recommendations; share final report
Export councils and trade-support bodies (e.g., Agricultural Export Council, Food Export Council)	Primary user / influencer	Sector statistics for members and reports, ability to filter and export views	High	High	Use their published figures as cross-checks; offer exportable views
Date farmers, producers and cooperatives	Secondary user	Demand signals, price trends, which varieties and markets are growing	Low	High	Plain-language insights; variety view if data allows

Stakeholder	Type	Needs and expectations	Influence	Interest	How we engage
Investors, packers and processors	Secondary user	Growth rates, market concentration and risk, opportunity assessment	Medium	Medium	CAGR, HHI and scenarios clearly explained
Data providers (UN Comtrade, ITC Trade Map, FAOSTAT; national statistics and customs bodies)	Data supplier	Correct citation and respect of terms of use	Medium	Low	Cite every source; record licences in the Sources Register
Importers and foreign buyers	Indirect	Reliable view of Egyptian supply and trends	Low	Medium	Consider English labels and clear definitions
Researchers, students and analysts	Secondary user	Transparent method, reusable data definitions	Low	Medium	Methodology page, data dictionary, GitHub repository
DEPI instructors and evaluation committee	Evaluator	Complete documents, working dashboards, clear contribution of each member, GitHub evidence	High	High	Deliver every milestone; keep documentation and commit history clear
Project team (BIS, MIS, 2 × GIS, Agriculture)	Builder	Clear scope, realistic plan, reliable data	High	High	Weekly meetings, shared task board, pull-request reviews

Influence / interest summary

Manage closely (high influence, high interest)
Ministry of Agriculture · export councils · DEPI evaluators · project team

Keep informed (lower influence, high interest)
Dates exporters · farmers and cooperatives · investors and processors

Keep satisfied (medium influence, lower interest)
Data providers

Monitor (lower influence and interest)
Importers · researchers and students

3. User Personas (illustrative)

Persona	Goal	Key questions
Export manager at a dates company	Decide which markets and products to focus on	Where is demand growing? Who am I competing with? What price level is realistic?
Policy analyst (ministry or export council)	Monitor national performance and dependency risk	How fast are exports growing? Are we too reliant on a few markets? What may 2026–2027 look like?
Investor / agribusiness analyst	Assess attractiveness of dates exports	What is the growth rate and volatility? What are the risks?

4. User Stories

ID	As a...	I want to...	So that...	Acceptance criteria	Priority
US-01	Exporter	see how total dates exports changed from 2016 to 2025	I understand the momentum of the sector	Line chart + YoY %; values match the source data	Must
US-02	Policy analyst	see KPIs for the latest year (value, YoY, CAGR, share of HS 0804)	I can read the situation in seconds	Four KPI cards on the overview page; definitions in tooltips	Must
US-03	Any user	filter by year range and product	I can focus on what matters to me	Filters update all views on the page within 3 s	Must

ID	As a...	I want to...	So that...	Acceptance criteria	Priority
US-04	Policy analyst	compare dates with other HS 0804 products	I can see the role of dates in the category	Stacked/100% chart and share table	Must
US-05	Researcher	read the data sources, definitions and limitations	I can trust and reuse the analysis	Methodology page lists sources, units, dates, limitations	Must
US-06	Exporter	see the top importing markets and their share Conditional	I can choose where to focus	Ranked top-10 list or map with shares; year filter	Should
US-07	Policy analyst	see market concentration (HHI) Conditional	I can judge dependency on a few markets	HHI value and explanation for the selected year	Should
US-08	Exporter	compare Egypt with competing exporters Conditional	I know where Egypt stands	Value, share and growth for Egypt vs main competitors	Should
US-09	Exporter	see average price per tonne over time Conditional	I can benchmark my price	Price = value ÷ quantity; trend line; units stated	Should
US-10	Policy analyst	view 2026–2027 scenarios	I can plan support measures	Low / base / high scenarios with assumptions and error range	Should
US-11	Export council analyst	export a view as image or PDF	I can use it in my own reports	Export option available; layout stays readable	Should
US-12	Any user	read a short list of key insights and recommendations	I get the message in 30 seconds	≥ 8 insights with figure, source and implication	Should
US-13	Farmer or exporter	compare date varieties Conditional	I know which varieties sell best	Variety comparison with a cited data source	Could
US-14	Researcher	see the main obstacles to Egyptian dates exports	I understand what limits growth	Short summary with cited studies (optional/advanced)	Could
US-15	DEPI evaluator	open the GitHub repository with documents, code and dashboard link	I can assess completeness and each member's work	README, folder structure, commit history, deployed link	Must

5. Use Cases

5.1 Use case list

ID	Use case	Primary actor	Related stories	Priority
UC-01	View executive overview	Exporter, policy analyst, investor	US-01, 02, 12	Must
UC-02	Filter and explore trends and product mix	Any user	US-03, 04	Must
UC-03	Analyse markets and concentration Conditional	Exporter, policy analyst	US-06, 07	Should
UC-04	Benchmark against competitors Conditional	Exporter, investor	US-08, 09	Should
UC-05	Compare date varieties Conditional	Exporter, farmer	US-13	Could
UC-06	View forecast scenarios	Policy analyst, investor	US-10	Should
UC-07	Read methodology and data sources	Researcher, any user	US-05	Must
UC-08	Export or share a view	Export council analyst	US-11	Should
UC-09	Refresh and validate data	Data analyst (team / maintainer)	Supports all	Must

5.2 Key use case descriptions

Field	UC-01 – View executive overview
Actor	Exporter, policy analyst or investor
Preconditions	Dashboard published; data validated
Main flow	1. User opens the dashboard. 2. System shows KPI cards (latest-year value, YoY, CAGR, share of HS 0804) and the 2016–2025 trend. 3. User hovers a point to read the value and definition. 4. User reads the key-insight panel.
Alternative / exception	If the data fails validation, the page shows a data-status warning instead of KPIs.
Postcondition	User understands the headline performance of dates exports.

Field	UC-03 – Analyse markets and concentration
Actor	Exporter or policy analyst
Preconditions	Country-level export data loaded and validated

Field	UC-03 – Analyse markets and concentration
Main flow	1. User opens the Markets page. 2. System shows the top importing countries (map and ranked bars) with share. 3. User selects a year. 4. System updates shares and the HHI value.
Alternative / exception	If country-level data is missing for the selected year, the page states the limitation.
Postcondition	User identifies main markets and dependency risk.

Field	UC-06 – View forecast scenarios
Actor	Policy analyst or investor
Preconditions	Historical series validated; forecast method documented
Main flow	1. User opens the Forecast page. 2. System shows history and low/base/high scenarios for 2026–2027. 3. User reads assumptions and error range.
Alternative / exception	If the forecast fails its back-test threshold, the page labels it “indicative only”.
Postcondition	User understands plausible export paths and their uncertainty.

Field	UC-09 – Refresh and validate data
Actor	Data analyst
Preconditions	New or corrected source data available
Main flow	1. Analyst places the raw file in data/raw. 2. Pipeline transforms the data (units, HS level, country names). 3. Validation checks run (totals reconcile, no missing years, no negatives). 4. Analyst reloads the Tableau extract and republishes.
Alternative / exception	If a check fails, the load stops and the issue is logged.
Postcondition	Dashboard shows validated data with an updated source note.

6. Functional Requirements

ID	Name	Requirement	Priority	Source	Note
FR-01	Data import	The solution ingests the cleaned export datasets (2016–2025) into the data model.	Must	US-01	
FR-02	Data validation	Data passes validation checks (HS-level totals reconcile, no missing years, no negative values) before use in dashboards.	Must	UC-09	
FR-03	KPI cards	Overview page shows latest-year export value, YoY growth, CAGR and share of HS 0804.	Must	US-02	
FR-04	Trend view	Line chart of annual export value 2016–2025 with YoY change.	Must	US-01	
FR-05	Filters	Filters for year range and product; country filter where data exists.	Must	US-03	
FR-06	Product mix	Composition of HS 0804 by product, with the share of dates and a ranked table.	Must	US-04	
FR-07	Growth diagnostics	YoY changes with annotations for key events, each backed by a cited source.	Should	US-01	
FR-08	Market analysis	Top importing countries (map and ranked bars) with share and HHI.	Should	US-06, 07	Conditional
FR-09	Competitor benchmarking	Egypt vs main exporters: value, world share and growth.	Should	US-08	Conditional
FR-10	Price analysis	Average export price per tonne over time, with units stated.	Should	US-09	Conditional
FR-11	Variety comparison	Compare date varieties by export or sales contribution.	Could	US-13	Conditional
FR-12	Forecast scenarios	Low / base / high scenarios for 2026–2027 with assumptions and error range.	Should	US-10	
FR-13	Insights panel	At least 8 documented insights and recommendations, each with figure, source and implication.	Should	US-12	
FR-14	Interactivity	Tooltips with definitions, highlight and drill-down actions.	Should	US-03	
FR-15	Methodology page	Lists data sources, units, definitions, update dates and limitations.	Must	US-05	
FR-16	Export / share	Users can export a view as image or PDF.	Should	US-11	
FR-17	Navigation	Consistent navigation between all dashboard pages.	Must	US-01–04	
FR-18	Obstacles summary	Short summary of the main obstacles to Egyptian dates exports, from cited studies.	Could	US-14	Optional
FR-19	Repository & docs	Code, data (where permitted), documents and the dashboard link are in GitHub with a README.	Must	US-15	

7. Non-functional Requirements

ID	Category	Requirement	How verified
NFR-01	Performance	Each dashboard page renders in ≤ 5 s and filters respond in ≤ 3 s on a normal broadband connection.	Timed tests
NFR-02	Accuracy & integrity	All displayed totals reconcile to the source; 100% of validation checks pass before publishing.	Validation report
NFR-03	Traceability	Every metric has a documented definition and source; no figure appears without one.	Data dictionary audit
NFR-04	Usability	A first-time user finds the top market and the growth trend in ≤ 2 minutes without training; UAT satisfaction ≥ 80%.	UAT with ≥ 5 users
NFR-05	Accessibility	Text contrast ≥ 4.5:1; meaning is never conveyed by colour alone; readable font sizes.	Design checklist
NFR-06	Compatibility	Works in current Chrome, Edge, Firefox and Safari, on desktop and tablet; mobile layout reviewed.	Cross-browser test
NFR-07	Reliability & availability	Dashboards reachable through a published link; data extracts and workbooks versioned so any version can be restored.	Link check, GitHub history
NFR-08	Security & privacy	Only public, aggregated data; no personal data; data licences and terms of use respected.	Sources Register review
NFR-09	Maintainability	Modular pipeline with a data dictionary and README; adding a new year takes ≤ 1 day.	Documented refresh test
NFR-10	Scalability	The star schema accepts new years, products and countries without redesign.	Schema review
NFR-11	Language	Dashboard labels and documents in English (project language); Arabic labels are a future option.	Design review

8. Assumptions, Constraints and Dependencies

Assumptions

- Public trade and production data can be downloaded and cited
- Tableau (Desktop or Public) is available to the team
- The team meets weekly

Constraints

- DEPI deadlines: 16 Oct, 6 Nov, 30 Nov, 4 Dec 2026
- Annual data, 10 years only (2016–2025)
- The provided file has world totals only and no varieties

> **Known data gap.** The export dataset does not identify date varieties, and has no importing countries or quantities. Requirements FR-08 to FR-11 are therefore **conditional**: they are built only if reliable data is found; otherwise the limitation is documented and the item moves to future work.

9. Traceability: Research Questions → Requirements

Research question	Requirements	Dashboard page
RQ1 Growth and its drivers	FR-03, 04, 07, 13	Overview · Growth diagnostics
RQ2 Share of dates in HS 0804	FR-06	Product mix
RQ3 Markets and competitors	FR-08, 09	Markets & competitors
RQ4 Price per tonne	FR-10	Markets & competitors
RQ5 Scenarios 2026–2027	FR-12	Forecast & recommendations
Extra: varieties · obstacles	FR-11 · FR-18	Product mix · Insights

10. Open Points

- Identify a reliable source for date varieties and prices (owner: Agriculture member).
- Confirm availability of country-level and quantity data from UN Comtrade / ITC Trade Map (owner: GIS member).
- Validate stakeholder needs with at least five real or representative users before the 30 Nov implementation milestone.